LESSON 9 • TERM 4 • LEVEL 1

The Problem-Solving Toolbox

Pick the tool, then check it

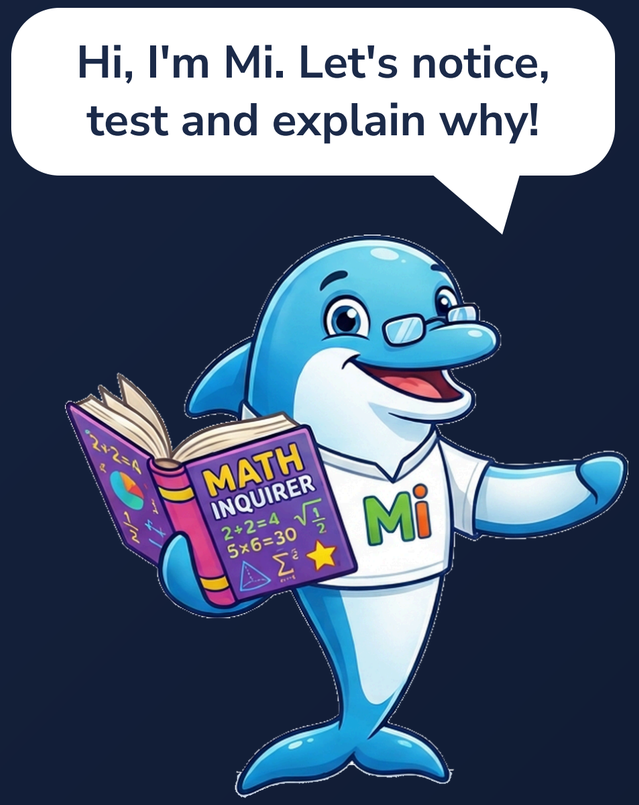


TODAY WE WILL

Meet seven tools.

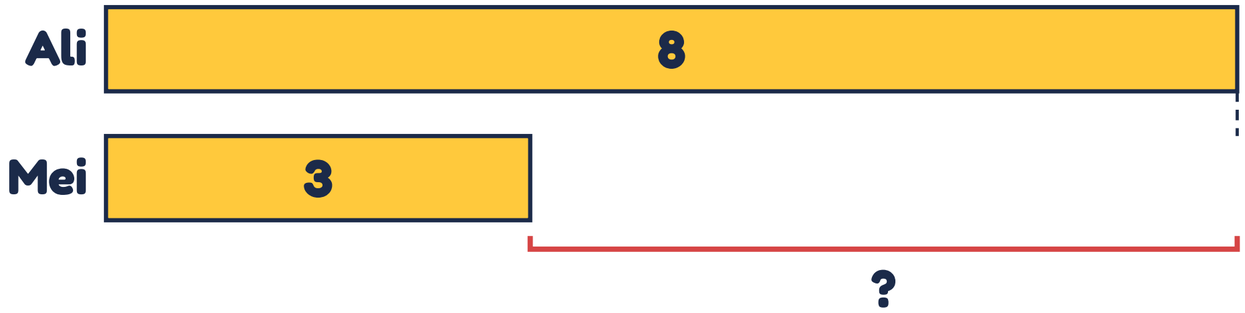
Pick the tool that fits.

Check an answer a second way.



Warm-up: draw two strips

2–8 MIN



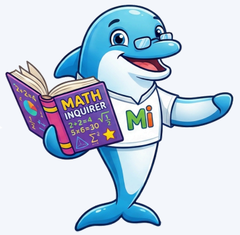
Ali has 8 stickers. Mei has 3 stickers. How many more stickers does Ali have than Mei?

Find one first

2–8 MIN



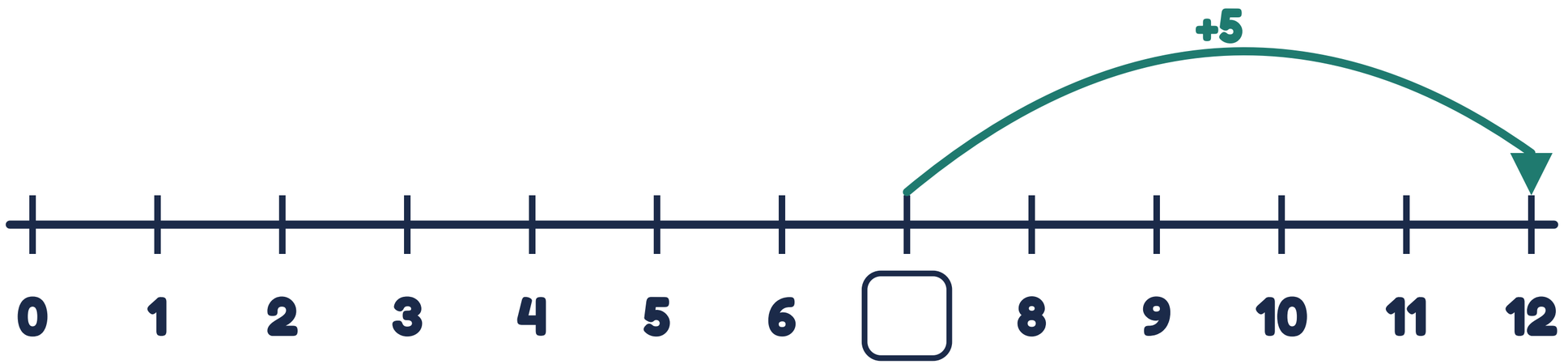
3 pens cost $12. All the pens cost the same. How much does 1 pen cost?



Take your time!

Work backwards

2–8 MIN



I think of a number. I add 5. I get 12. What is my number?

Sneak-peek puzzles

8–12 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

PUZZLE 1: anyone can try

Mei has 4 stickers.

Ali has 3 more stickers than Mei.

How many stickers does Ali have?

PUZZLE 2: think

Mei and Ali have 10 stickers in all.

Ali has 2 more stickers than Mei.

How many stickers does Mei have?

I know because ___.

Explore: your seven tools

12–23 MIN

Strip	Table	List	Draw it
Work backwards	Find one first	Steps	

Lay your 7 tool cards in a row. Which tool have you used before?

Explore: problems 1 to 4

12–23 MIN

Card	Problem
1	Ali has 9 pens. Mei has 4 fewer pens than Ali. How many pens do they have in all?
2	Rows of blocks have 2, 4, 6 blocks, and so on. Each row has 2 more than the row before. How many blocks are in the 6th row?
3	Mei picks 2 fruits from an apple, a pear and a plum. List every pair she can pick.
4	Ravi is 3rd in a line of 7 children. How many children are behind Ravi?

Put each problem card under a tool you would use. Say why.

12–23 MIN

A tool that fits: 1 to 4

Card	A tool that fits
1	Strip
2	Table
3	List
4	Draw it

More than one tool can fit. A good reason matters more than the name.

Explore: problems 5 to 8

12–23 MIN

Card	Problem
5	A number take away 6 is 9. What is the number?
6	5 buns cost $15. All the buns cost the same. How much do 8 buns cost?
7	Ben has 20 sweets. He eats 8 sweets. Then he buys 5 more sweets. How many sweets does he have now?
8	For every 2 red beads there are 3 blue beads. How many blue beads go with 8 red beads?

Put each problem card under a tool you would use. Say why.

12–23 MIN

A tool that fits: 5 to 8

Card	A tool that fits
5	Work backwards
6	Find one first
7	Steps
8	Table

Problems 2 and 8 both use a table. Many tools can fit one problem.

What did we notice?

12–23 MIN

- Does one word, like more, tell us the tool?
- Can two tools fit the same problem?

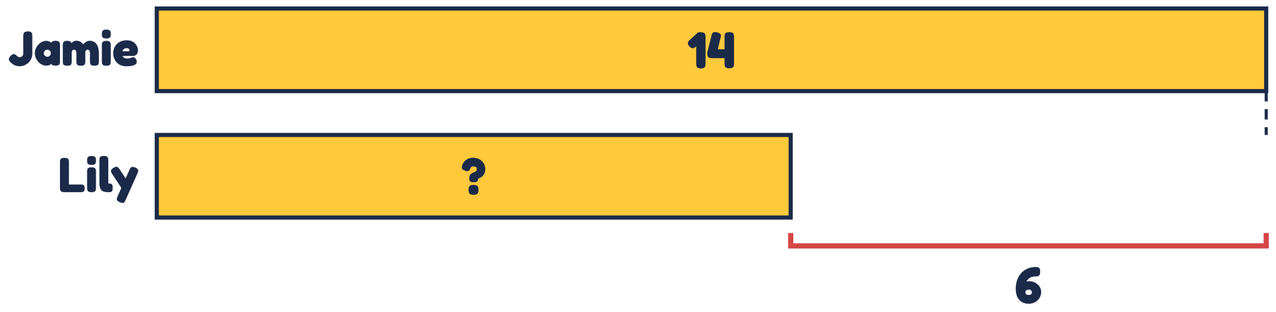
Seven tools, seven clues

23–35 MIN

	Tool	Use it when
	Strip	Compare, or find a total.
	Table	Numbers grow in a pattern.
	List	Find every way.
	Draw it	A line, a place or a shape.
	Work backwards	You know the end, not the start.
	Find one first	Equal groups, or the price of one.
	Steps	One thing happens, then the next.

Strip: more and fewer

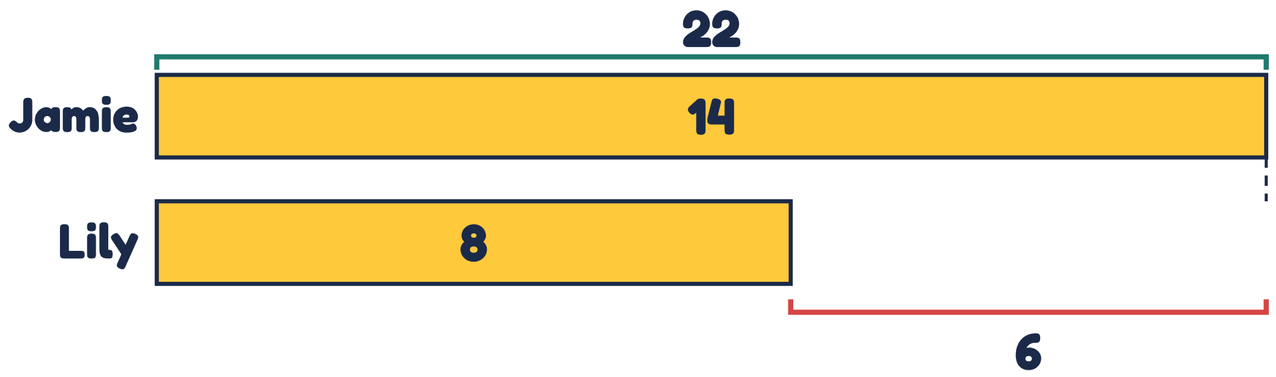
23–35 MIN



Jamie has 14 cards. Lily has 6 fewer cards than Jamie. How many cards do they have in all?

23–35 MIN

Lily has 8 cards, 22 in all



14 take away 6 is 8 cards for Lily. 14 + 8 = 22 cards in all.

Table: a growing pattern

23–35 MIN

Path	1	2	3	4	5	6	7
Tiles	2	5	8				

A path has 2 tiles, then 5 tiles, then 8 tiles. Each path has 3 more tiles than the path before. How many tiles are in the 7th path?

20 tiles in the 7th path

23–35 MIN

Path	1	2	3	4	5	6	7
Tiles	2	5	8	11	14	17	20

Add 3 each time: 2, 5, 8, 11, 14, 17, 20. The 7th path has 20 tiles.

Work backwards

23–35 MIN

?	+ 9		− 4	20

I think of a number. I add 9. Then I take away 4. I get 20. What is my number?

My number is 15

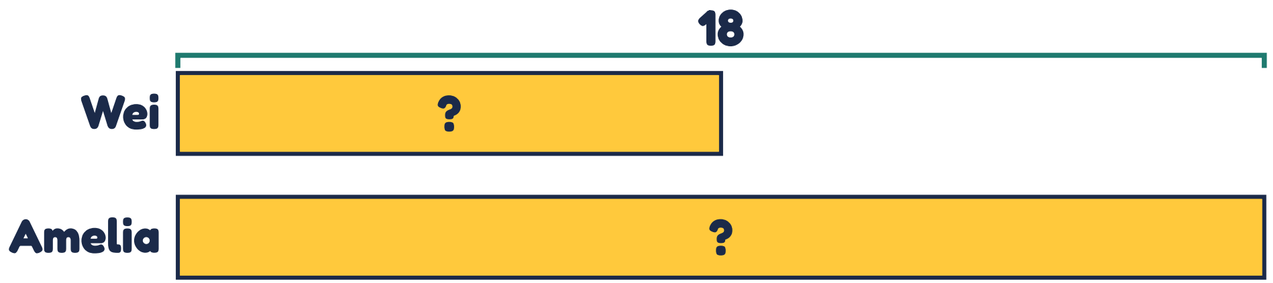
23–35 MIN

15	+ 9	24	− 4	20

Start at 20 and undo each step: 20 + 4 = 24, then 24 take away 9 is 15. Check: 15 + 9 = 24 and 24 take away 4 is 20.

Two roads, one answer

23–35 MIN



Amelia has twice as many marbles as Wei. They have 18 marbles in all. How many marbles does Wei have?

Road 2: a table of tries

23–35 MIN

Wei	Amelia	In all
4	8	12
5	10	15
6	12	18

Road 1, the strip: 3 equal parts, 18 shared into 3 is 6. Road 2, the table: Wei 6 and Amelia 12 make 18. Both roads say Wei has 6 marbles.

35–39 MIN

Mime the tools

Stand up. Mime each tool with me.

Draw a long strip in the air. Tap a table.

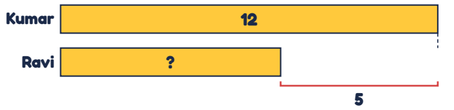
Sip some water.

Workbook page 1: Start here

39–70 MIN

Questions 1 to 3. Take your time.

S1 Use a strip. Kumar has 12 marbles. Ravi has 5 fewer marbles than Kumar.



S2 Do one step at a time. A shop has 30 apples. It sells 12 apples in the morning and 9 apples in the afternoon.

S3 Make a table. Each row has 3 more beads than the row before. Fill in the table.

Row	1	2	3	4	5	6
Beads	3	6	9			

39–70 MIN

Check page 1

S1	7 marbles; 19 marbles in all
S2	18 apples; 9 apples
S3	15 beads; 18 beads

Workbook page 2: Start here

39–70 MIN

Questions 4 to 6. Take your time.

S4 Work backwards. Ravi thinks of a number. He takes away 8 and gets 17.

S5 Find one first. 4 bags hold 20 apples. All the bags hold the same number of apples.



S6 Use a list. Ali has three coins: 5¢, 10¢ and 20¢. He picks 2 coins. Fill in the blanks.



PLAY MONEY

39–70 MIN

Check page 2

S4	25; 17
S5	5 apples; 35 apples
S6	15¢, 25¢, 30¢; 3 pairs

39–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	The tool is not named: you choose, and say why
Go further (page 4)	Two Roads, One Answer, make three amounts equal, a pattern
Investigation (page 5)	Pick a tool, write a problem
Check	Finish and check your own work

Everyone may choose. Not finished? Pages 3 to 5 are for home too.

Talk and explain

70–76 MIN

A CHILD IN ANOTHER CLASS SAYS

Amelia has twice as many marbles as Wei.

They have 24 marbles in all.

24 shared into 2 is 12, so Wei has 12 marbles.

Do you agree? How could we check a second way?

I think … because …

Back to the sneak-peek

70–76 MIN

Puzzle 1: 7 stickers. Puzzle 2: Mei has 4 stickers. Mei 4 and Ali 6 make 10, and 6 is 2 more than 4.

Exit ticket

76–82 MIN

On your own. No looking back.

E1 **Mei thinks of a number. She adds 4 and gets 15.**

Circle the tool that fits best.

E2 **Ali has 11 stickers. Mei has 4 fewer stickers than Ali.**

(a) Mei has ☐ stickers.

(b) They have ☐ stickers in all.

E3 **5 buns cost $20. A child in another class says 1 bun costs $5.**

Circle True or False.

Check the exit ticket

76–82 MIN

E1	Work backwards
E2	7 stickers; 18 stickers
E3	False

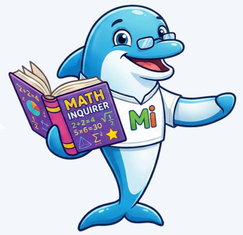
Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ picking a tool ☐ using a strip ☐ working backwards ☐ something else
What helped me	☐ the tool cards ☐ drawing it ☐ checking a second way ☐ I was not stuck
Next time I will try	☐ picking a tool ☐ checking a second way ☐ saying why ☐ asking a question
	☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

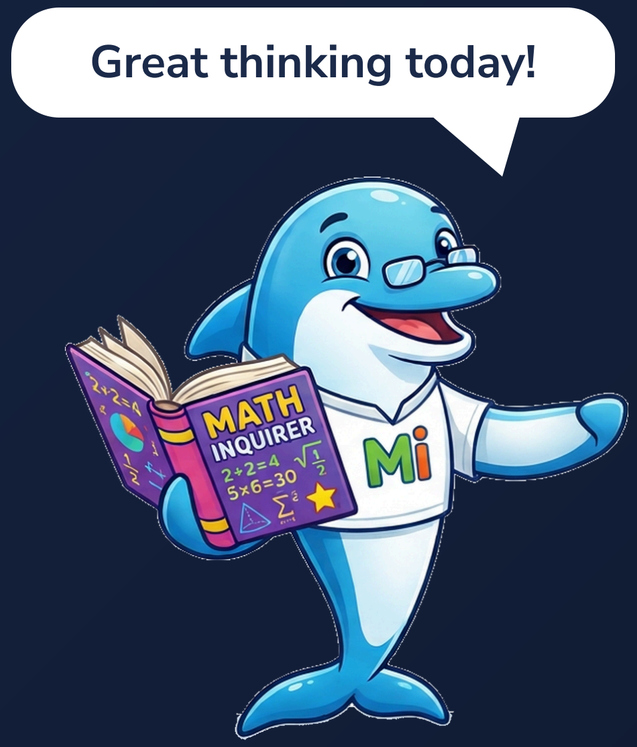
Well done, inquirers!

Show a grown-up your tool cards. Say which tool you like best.

NEXT LESSON

Inquirers' Showcase and Mi's Challenge Quest

Have ready: your Quest Log, closed in its envelope, a quest partner, a favourite page from your book and plain paper.



Go further: how many cuts?

OPTIONAL

A rope is cut into 5 pieces. How many cuts are there? Draw it.

OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	Can two different tools solve the same problem? Try one.
Make a puzzle	Pick a tool card and write a problem for your grown-up.
Teach it back	Tell a toy why we check an answer a second way.

Optional. Everyone may join.